

Jeongsang Ryu

PhD Candidate, Mechanical Engineering

UNIST, High-Assurance Mobility and Control Lab

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Research Interests

SLAM, LiDAR-Inertial Odometry, Autonomous Racing, End-to-End Autonomous Driving, Reinforcement Learning, Robot Perception

Education

Mar 2024 – **Combined MS/PhD, Mechanical Engineering, UNIST**
Feb 2030 High-Assurance Mobility and Control Lab (HMCL)
(expected) Research: SLAM, LiDAR-Inertial Odometry, RL for Autonomous Racing
Mar 2018 – **BS, Mechanical Engineering, UNIST**, GPA: 3.75 / 4.3
Feb 2024

Experience

Spring 2024, 2025, 2026 **Teaching Assistant — MEN 4xx: Creating Autonomous Car, UNIST, Mechanical Engineering**
Aug 2023 – **Undergraduate Research Intern, High-Assurance Mobility and Control Lab, UNIST**
Feb 2024 Autonomous racing algorithm development on F1TENTH platform. Motion planning, control, and ROS-based simulation and real vehicle testing.
Mar 2021 – **Undergraduate Research Intern, Clean Combustion & Energy Research Lab, UNIST**
Nov 2022 Neural network models for chemical kinetics simulation. Flame stabilization and extinction research using ML (Python, TensorFlow, OpenFOAM).
Apr 2019 – **Military Service, Republic of Korea Army**
Nov 2020

Skills

Languages C++, Python, Swift
Frameworks ROS / ROS2, Isaac Sim, Isaac Lab, ARKit, PyTorch, Docker
Platforms Linux, iOS

Awards

To be updated.

Publications

To be updated.